

CHOIBALSAN, Mongolia

WMO: 442590

Lat: 48.0811N

Lon: 114.5375E

Elev: 2451

StdP: 13.44

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-19.7	-30.5	1.1	-22.6	-26.7	1.4	-18.7	22.0	2.8	19.9	0.3	6.5	270	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	91.8	63.1	87.7	62.0	84.0	61.3	67.5	81.5	65.9	78.5	64.5	76.5	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	96.6	71.1	62.0	91.0	70.2	60.3	85.7	69.2	33.5	81.6	32.2	78.2	31.0	76.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	23.1	20.5	17.9	DB	-26.4	97.5	5.2	4.9	-30.1	101.1	-33.2	103.9	-36.1	106.7	-39.8	110.3
(5)				WB	-27.0	70.2	5.3	1.9	-30.8	71.6	-33.9	72.7	-36.9	73.7	-40.8	75.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	-4.3	4.2	20.4	40.1	54.0	66.0	71.1	66.8	54.7	37.0	15.3	0.5	(6)
(7)		DBStd	27.90	8.93	10.03	11.71	9.29	8.59	7.52	5.81	6.45	8.46	9.55	11.98	9.11	(7)
(8)		HDD50	7289	1683	1281	916	319	55	1	0	0	45	413	1042	1534	(8)
(9)		HDD65	11153	2148	1701	1381	748	355	76	10	54	322	867	1492	1999	(9)
(10)		CDD50	2057	0	0	0	22	179	480	654	523	188	11	0	0	(10)
(11)		CDD65	445	0	0	0	0	14	106	199	111	15	0	0	0	(11)
(12)		CDH74	4880	0	0	0	21	284	1297	1931	1140	205	2	0	0	(12)
(13)		CDH80	1842	0	0	0	3	91	508	768	419	53	0	0	0	(13)
(14)	Wind	WSAvg	7.8	7.6	7.6	8.3	9.9	9.8	7.6	6.7	6.4	7.4	7.6	7.4	7.7	(14)
(15)	Precipitation	PrecAvg	3.9	0.5	0.4	0.3	0.1	0.2	0.3	0.4	0.9	0.5	0.2	0.3	0.2	(15)
(16)		PrecMax	11.4	4.2	3.9	2.0	0.9	1.2	2.6	4.0	5.6	4.2	1.5	2.5	1.5	(16)
(17)		PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.5	0.9	0.7	0.5	0.2	0.3	0.7	0.8	1.2	0.8	0.3	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	37.1	56.8	77.2	89.0	96.8	98.4	94.1	86.2	70.7	48.4	27.4	(19)
(20)		0.4%	MCWB	19.4	29.5	39.7	49.7	56.2	62.1	66.8	64.7	59.2	50.0	36.9	22.5	(20)
(21)		2%	DB	16.9	29.2	49.6	68.1	81.6	90.5	92.0	88.8	80.0	62.3	42.0	21.7	(21)
(22)		2%	MCWB	14.3	24.3	36.1	46.4	53.9	61.2	64.7	62.6	56.7	45.5	33.4	18.4	(22)
(23)		5%	DB	12.7	24.1	43.1	61.9	76.0	85.7	87.7	84.5	74.8	56.7	36.7	17.3	(23)
(24)		5%	MCWB	10.7	20.4	32.8	43.0	52.3	60.0	64.2	61.5	55.2	42.5	29.8	14.8	(24)
(25)		10%	DB	8.7	19.5	37.8	57.0	71.3	81.3	83.7	80.3	70.0	51.8	32.1	13.4	(25)
(26)		10%	MCWB	7.1	16.7	29.6	40.8	50.2	59.0	63.4	61.0	53.5	40.1	26.9	11.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	29.5	41.2	50.2	59.0	65.6	70.5	68.7	63.4	50.5	37.5	23.3	(27)
(28)		0.4%	MCDWB	22.2	36.8	55.4	73.3	83.1	82.6	86.2	85.0	78.9	68.6	47.4	26.7	(28)
(29)		2%	WB	14.7	24.7	36.7	46.8	55.9	63.8	68.5	66.7	59.8	46.6	33.8	18.6	(29)
(30)		2%	MCDWB	17.0	29.3	48.9	66.6	76.3	80.5	83.2	78.7	71.2	59.9	42.1	21.8	(30)
(31)		5%	WB	11.0	20.7	33.1	43.7	53.5	62.4	67.0	65.0	57.5	43.6	30.4	15.0	(31)
(32)		5%	MCDWB	13.0	24.3	42.6	60.5	73.1	79.2	80.6	75.9	70.5	55.3	36.7	17.4	(32)
(33)		10%	WB	7.4	17.1	30.0	41.0	51.2	61.0	65.6	63.5	55.1	40.9	27.4	11.6	(33)
(34)		10%	MCDWB	9.1	19.9	37.4	56.2	69.1	76.4	78.1	74.4	68.0	51.2	32.4	13.4	(34)
(35)	Mean Daily Temperature Range		MDBR	16.0	19.8	20.5	22.5	24.2	21.9	19.8	20.9	22.3	20.2	18.0	15.2	(35)
(36)		5% DB	MCDBR	20.1	24.3	26.5	30.9	31.9	28.6	26.3	28.0	29.0	27.4	22.7	18.8	(36)
(37)			MCWBR	18.0	19.9	16.5	16.3	14.3	10.5	8.5	9.8	13.1	15.6	15.8	15.9	(37)
(38)			MCDBR	19.9	24.0	25.5	29.5	28.9	23.3	21.1	20.2	24.1	25.2	22.2	18.2	(38)
(39)		5% WB	MCWBR	17.8	20.0	16.6	16.0	13.9	9.2	8.4	8.9	11.9	15.3	15.8	15.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.278	0.319	0.387	0.456	0.458	0.422	0.382	0.337	0.308	0.275	0.260	(40)	
(41)		taud	2.291	2.259	2.236	2.106	1.949	2.063	2.241	2.325	2.386	2.384	2.343	2.261	(41)	
(42)		Ebn at Noon	262	280	284	272	256	257	265	272	275	266	252	242	(42)	
(43)		Edh at Noon	23	30	36	46	56	51	42	37	31	26	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	605	993	1444	1790	2033	2056	1940	1783	1443	994	628	459	(44)	
(45)		RadStd	43	55	68	90	99	137	103	103	75	55	32	46	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-3.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page